Ahmed J O. Ahmed

Generative AI Engineer | Agentic AI | LLM Systems | AWS Certified
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PROFESSIONAL SUMMARY

Generative AI Engineer with 3+ years of experience architecting and deploying production-grade LLM applications, multi-agent systems, and RAG pipelines using LangChain, LangGraph, CrewAI, and modern orchestration frameworks. Expert in LLM fine-tuning, prompt engineering, vector search, and agentic AI workflows for intelligent automation and decision-making. Proven track record delivering AI-powered solutions across healthcare, security, and enterprise automation, with a focus on responsible AI, hallucination mitigation, and scalable GenAI architectures.

TECHNICAL SKILLS

Languages: Python (4+ years), MATLAB, SQL
ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, LightGBM, Hugging Face Transformers, Ultralytics, CUDA, Transformers, Vision Transformers (ViT), LSTM, CNNs, YOLO, Whisper, Grad-CAM
GenAI & LLMs: LLM fine-tuning (LoRA/QLoRA), RAG, Graph RAG, Prompt Engineering, Vector Search (FAISS, Pinecone, Chroma), Knowledge Graphs, VLMs (CLIP, BLIP), Hallucination Detection, INT4 Quantization, ONNX export, Knowledge Distillation
Agentic AI & Orchestration: LangChain, LangGraph, CrewAI, GPT-4, Mistral AI — multi-agent orchestration, role-based task decomposition, tool-use agents, automated reasoning pipelines, feedback-driven learning loops
Cloud & MLOps: AWS (SageMaker, Lambda, EC2, S3, ECS, ECR, RDS, CloudWatch, Athena, EMR), Docker, FastAPI, GitHub Actions (CI/CD), MLflow, Redis, Celery, Postgres/pgvector, MongoDB
Data & Visualisation: Pandas, NumPy, OpenCV, spaCy, NLTK, MongoDB, Neo4j, Power BI, Plotly, Streamlit, Matplotlib, Seaborn
Edge AI & Leadership: Raspberry Pi, NVIDIA Jetson Nano, real-time model optimisation; end-to-end project ownership, cross-functional collaboration, conference presentations, mentorship

EXPERIENCE

Data Scientist | Natural History Museum – London *Aug 2026 — Present*

- Contributing to SCRIBE (Structured Collection Records Interpretation and Bio-entity Extraction), an AI-driven project within the Museum's AI & Innovation team, aimed at digitising and structuring millions of historic natural science collection records and manuscripts as part of a UK-wide effort to unlock the national natural science collections.
- Built an OCR pipeline using Vision-Language Models (VLMs) to extract structured, queryable data directly from scanned and handwritten historic specimen labels and manuscripts, converting unstructured document images into standardized structured output.
- Applied a broad AI/ML stack spanning computer vision, NLP, and LLM-based extraction to process collection data at scale, supporting the transition from an early proof-of-concept to a production-ready, scalable digitisation platform.

AI Engineer | Wyxan – Remote *Jun 2025 — Aug 2026*

- Architected an LLM-powered “Security Guard Owl” agentic system using generative AI to automatically produce professional, translated, standardized incident reports, reducing report defects by 25%.
- Integrated Pixtral (Mistral AI) Large as a vision-language agent to verify reports against images, implementing multimodal reasoning for real-time consistency validation.
- Designed feedback-driven anomaly detection and report-consistency agents with supervisor dashboards, helping cut incidents by 30% over six months through autonomous monitoring.

AI Engineer | Ayoob AI – United Kingdom *Mar 2025 — Jun 2025*

- Engineered generative AI automation pipelines for logistics and retail using LLM-driven workflows, reducing manual effort by 40%; deployed via AWS and Docker.
- Prototyped agentic AI tools with LangChain and GPT-4, designing privacy-preserving LLM orchestration pipelines to keep sensitive business data confidential during multi-step reasoning.

AI Research Engineer | Sudan University of Science and Technology – Sudan *2023 — 2024*

- Conducted applied AI research combining computer vision with automated decision systems, leading to a published IEEE ICCA 2023 paper on real-time plant disease detection.
- Developed and deployed YOLOv4-based monitoring pipelines on Raspberry Pi for field environments.

Trainee Engineer | Sudanese Electricity Distribution Company – Sudan *Jan 2018 — Dec 2020*

- Supported data-driven optimization of power systems and renewable energy operations.
- Applied core engineering principles in internship projects focused on energy efficiency and reliability.

KEY PROJECTS

VidInstructQA — Training-Free Video Question Answering — *BlackboxNLP @ EMNLP 2026* | *Link*

- Developed a training-free VideoQA pipeline pairing a frozen LLaVA-NeXT-Video-7B generator with a LLaMA-3.1-8B LLM-as-judge evaluator, demonstrating advanced prompt engineering across five exemplar sets on NExT-QA and MSVD-QA.
- Quantified exemplar-choice variance dominating prompting condition (18-point one-shot swing on NExT-QA), showing that single-exemplar reports overstate typical gains.
- Diagnosed a label-bias failure mode in LLM-as-judge evaluation, where prompt demonstration bias inflated accuracy by 27 points; validated with human annotators and Claude Opus 4.7 as tiebreaker (κ=0.693/0.695).

GraphMind — Knowledge Graph QA System

- Architected a Graph RAG knowledge QA system on Neo4j with LangGraph-orchestrated retrieval agents, improving factual consistency by 35% over baseline LLM retrieval.
- Designed a multi-agent report generation system (Research, Analysis, Report Agents) with role-based task decomposition, reducing manual effort by 60%; implemented hallucination detection metrics and expert feedback loops.
- Built a Streamlit dashboard for real-time agentic querying and knowledge graph visualization across enterprise data sources.

MeetingMindr — Multi-Modal Meeting Summarization — *Link*

- Developed a multi-modal AI pipeline combining Whisper/PyAnnote transcription with GPT-4o/Gemini LLM summarization agents, reducing manual note-taking by 90%.
- Engineered a privacy-preserving LLM architecture for proprietary meeting data, with an automated QA agent validating factual accuracy and hallucination-free completeness.
- Integrated LLM output into collaboration tools with timestamped action items for seamless follow-up.

Medical Imaging with Vision Transformers — *Link*

- Applied Vision Transformers and CNNs to tumor classification, achieving 70% accuracy with Grad-CAM explainability for clinical validation and regulatory compliance.
- Implemented rigorous validation including cross-validation and bias detection across demographic cohorts.
- Identified and fixed dataset leakage, improving model generalization on unseen patient data.

VisionAid — Smart Navigation Glasses for the Visually Impaired — *Workshop Paper* | *Demo*

- Real-time obstacle detection and audio feedback system using YOLOv5 deployed on NVIDIA Jetson Nano with mixed-precision inference and threaded I/O.
- Achieved >85% accuracy across object detection and sound localization; submitted workshop paper documenting system design and evaluation.

PUBLICATIONS

1. **Your Judge Inherits Your Prompt: In-Context Label Bias in an LLM-as-Judge Is Directional, Not Symmetric**
   Abdullah Ahmad, **Ahmed J. O. Ahmed,** Luca Citi
   Accepted, BlackboxNLP Workshop, EMNLP 2026
2. **Safety-Prioritised Multimodal Navigation for Edge Smart Glasses with Constrained On-Device Language Models** | Paper
   Sefki Kolozali, **Ahmed J. O. Ahmed,** Abdullah Ahmad, Alireza Kalantarhormozi, Soran Ghaderi, Mustafa Mustafa, Ali Radwan, Vishwanathan Mohan, Klaus McDonald-Maier
   International Digital Public Health Conference Workshop (DPH 2026)
3. **Real-Time Agricultural Monitoring with Agrobot: A Raspberry Pi and YOLO Based Solution** | Paper
   Ahmed J. O. Ahmed, A. Babiker, A. Elhag, M. Drar
   IEEE International Conference on Computing, Control and Applications (ICCA), 2023

EDUCATION

MSc in Artificial Intelligence — University of Essex, UK *Oct 2023 – Oct 2024*
Distinction (4.0/4.0 CGPA)
Advisor: Dr. Luca Citi

BSc in Electrical Engineering — Sudan University of Science and Technology *2015 – May 2022*
GPA: 3.17/4.0

CERTIFICATIONS

AWS Certified Machine Learning Specialty | **Certified Artificial Intelligence Engineer, USAII**

AI Ethics Course — Turing College